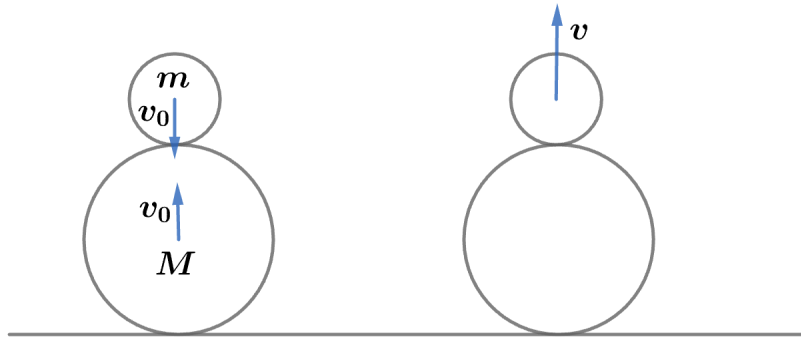


2007 F=ma Exam: Problem 16

Kevin S. Huang



By conservation of linear momentum (taking upward to be positive),

$$p_i = Mv_0 - mv_0 = mv = p_f$$

Solving for v ,

$$v = \frac{M - m}{m}v_0 = \frac{0.5 \text{ kg} - 0.2 \text{ kg}}{0.2 \text{ kg}}(4 \text{ m/s}) = 6 \text{ m/s}$$

so the answer is B.